



Sort Two Ways

The same things, grouped two ways

Name:

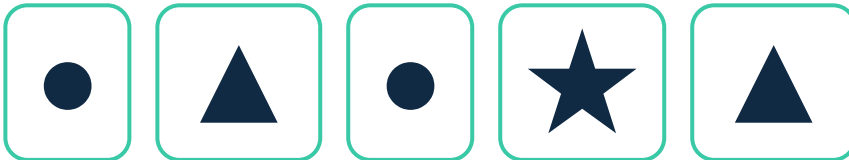
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★ I can sort the same shapes by more than one rule.

One pile, two ways to sort

The same shapes can be grouped in more than one way. It depends on the RULE you choose. Here is Bit's pile.

Bit's pile



1 Way 1 — sort by shape

Rule: same shape together. Draw the circles in one box, triangles in the next, the star in the last.

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2 Way 2 — round or pointy?

New rule: round shapes (●) in one box, pointy shapes (▲ ★) in the other. Sort the SAME pile again.

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3 You try — sort this pile two ways

Pile: ★ • ▲ •. First sort by shape. Then sort by round or pointy.

Big idea

The same things can be sorted in different ways — you just change the rule. Computers sort by different rules too: by size, by colour, by name.



Sort Two Ways

Quick facilitation notes & answer key

What this worksheet teaches

Children sort one pile two different ways — by exact shape, then by round-versus-pointy. Seeing the same set grouped different ways is a first thinking-skill behind coding — the rule decides the groups. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil or crayon. You read each prompt aloud; the child draws into the group boxes. Optional: a small pile of shape blocks to sort by hand two ways.

How to lead it (about 15 minutes)

1. Show them (you do). Name the pile — two circles, two triangles, one star.
2. Do one together. Way 1: sort by shape (Ex 1). Way 2: change the rule to round versus pointy (Ex 2) — circles are round; triangles and the star are pointy.
3. Let them try. Children sort the Exercise 3 pile both ways on their own.

Answer key (these are the verified correct answers)

Way 1 (by shape): circle = 2, triangle = 2, star = 1. Way 2 (round vs pointy): round = 2 (the circles); pointy = 3 (two triangles + one star).

Exercise 3 pile (star, circle, triangle, circle) — by shape: star 1, circle 2, triangle 1. Round/pointy: round 2, pointy 2.

If children get stuck — and ways to adjust

Watch for: re-sorting Way 2 by shape out of habit. Restate the new rule (round vs pointy).

Make it easier: do only Way 1 with hands-on blocks, then add the second rule.

Make it harder: invent a third rule together (big vs small, or has-a-point vs no-point).

Make it active: sort the children themselves two ways (e.g. by sleeve colour, then by shoe type).

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children classify the same objects by more than one attribute.

In plain terms: the same pile can be grouped different ways depending on the rule.

No reading is needed from the child — you read each prompt aloud; the child answers by drawing or circling.

You'll know it worked when

The child sorts the one pile correctly under both rules and can say that changing the rule changes the groups.